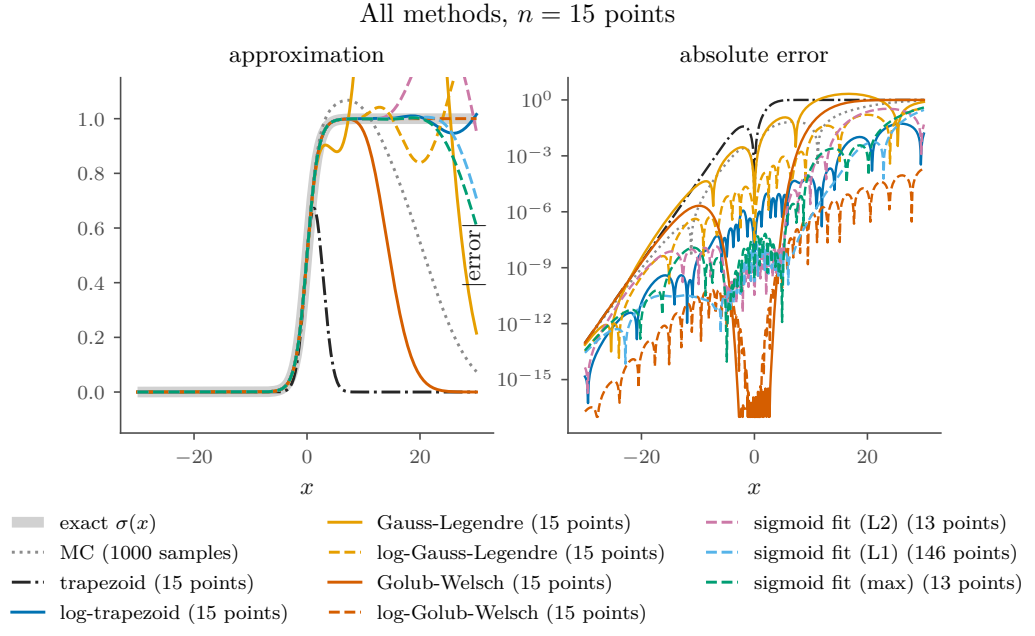

All methods, $n = 15$ points
approximation
absolute error
1.0
0.8
0.6
0.4
0.2
0.0
error
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-12}
10^{-15}
−20
0
20
x
exact $\sigma(x)$
MC (1000 samples)
trapezoid (15 points)
log-trapezoid (15 points)
Gauss-Legendre (15 points)
log-Gauss-Legendre (15 points)
Golub-Welsch (15 points)
log-Golub-Welsch (15 points)
sigmoid fit (L2) (13 points)
sigmoid fit (L1) (146 points)
sigmoid fit (max) (13 points)